

From Detention to Prevention: Using Education to Foster Safer Neighborhoods

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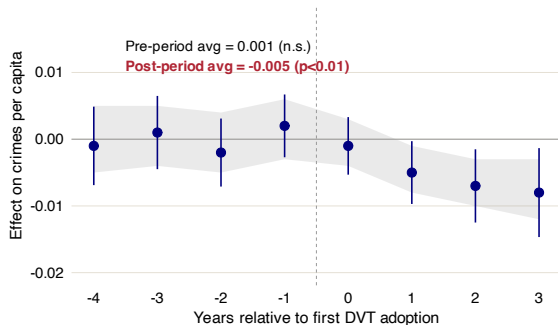
DVT cuts local crime $\sim 20\%$ at the mean adopter dose

Catalonia's 2014 Dual VT reform. Vocational students rebalance $\sim 400 \rightarrow \sim 1,000$ firm-based hours; degree, qualification, length unchanged. Staggered, decentralised rollout 2014/15 onward.

Design. Continuous-dose DiD on a 326-tract panel (2008/09–2017/18). Dose = share of male VT students in the dual track. Headline: dCDH; six robustness estimators overlaid.

Main results

- ▶ $ATT = -0.032$ (SE 0.016)
- ▶ $\approx 20\%$ at mean dose 0.
- ▶ Property crime, ages 14–33, foreign offenders
- ▶ No displacement to host-firm tracts



Stylised illustration of paper Fig. 3. Dots: dCDH; band: 5 alternative estimators.

What is exciting: a within-track, place-based test the literature has not had

- ▶ Most education-and-crime work studies academic schooling and compulsory-age shifters. Vocational education is understudied.
- ▶ **Within-track variation.** Degree and qualification are held fixed; only the classroom-vs-firm mix changes.
- ▶ Three exciting results:

Host-firm placebo
no displacement to firm sites

Residence aggregation
effect travels home with student

Across-the-clock
 $\text{weekend} \geq \text{weekday}$

- ▶ Together, these rule out pure incapacitation *and* pure school-neighborhood-vibe stories.
- ▶ **The mechanism story is very disciplined:** heterogeneity by age (Table 9), nationality (Table 7A), and time-of-day (Table 8) loads where a labour-market channel predicts.

Comment 1: This identifies the *delivery-format* effect, not the *paid-apprenticeship* effect

The Catalan implementation: 93% unpaid (EBCA), 7% paid contracts.

- ▶ **This is an identifying *advantage*.** The wage-income channel is muted by construction; the design isolates classroom-vs-firm time rebalancing without contamination from training wages.
- ▶ **But it weakens inference to contract-dominant systems** (Germany, Switzerland) where apprentices are paid from day one and the wage channel is large.
- ▶ **A policy maker may reach for “DVT works → expand paid apprenticeships”.** The design does not deliver that. It is a clean test of *delivery format*, a weak test of *paid apprenticeship*.

**Rethink the contribution: “the public-safety value of moving classroom hours into the firm,”
not “the public-safety value of an apprenticeship system.”**

Comment 2: A pre-trend on the variable that drives the headline

Table 2 regresses year of first DVT adoption on 2011–2013 changes. Most cells null. One is not.

Outcome (2011–2013 change), Panel A	Coef. (SE)
Crimes p.c. (all)	0.048 (0.035)
Crimes against the person p.c.	−0.082 (0.240)
Property crimes p.c.	−0.711* (0.366)

- ▶ **Sign:** earlier-adopting tracts were already experiencing *declines* in property crime.
- ▶ **Why this matters:** the headline reduction is concentrated in property crime (Table 6: theft, robbery, misappropriation drive the result).
- ▶ **The one timing-exogeneity cell that loads is on the exact outcome category that drives the headline,** with a sign that biases toward the result.

... and the pre-trend hints at bigger contemporaneous shocks

The Table 2 cell is local; the worry it points to is broader. DVT did not roll out in a vacuum — 2014/15 sits in a window crowded with other policy shocks.

- ▶ **2012 Spanish labor-market reform** — restructured youth employment contracts.
- ▶ **LOMCE education reform** (2013) — broader curricular and structural changes.
- ▶ **Rapidly declining youth unemployment** as Spain emerged from the crisis (≥ 2013).
- ▶ **Year FE absorb the *average* shock, not *differential* exposure.** If DVT-adopting tracts have different sectoral profiles than controls, exposure to these shocks differs systematically.
- ▶ **Concrete worry:** industrial-sector-heavy tracts — where DVT expanded fastest — were also where manufacturing employment recovered fastest after 2012.
- ▶ **Section 7.2's sectoral heterogeneity** is suggestive but not designed to isolate this confound.

**At minimum: discuss the contemporaneous policies. Better: control for municipal youth-unemployment trajectories,
or drop tracts dominated by sectors most exposed to the 2012 labor reform.**

Three avenues to push the paper further

Great and ambitious paper with rare administrative-data linkage and a solid identification strategy

The age-cohort heterogeneity (Table 9) is the strongest design-based evidence: null on every band 34+, significant on every band 14–33; joint $\chi^2(7) = 30.30$.

- 1 Spillover / SUTVA test.** Tracts are small ($\sim 1,700$) and contiguous. If treated students offend in adjacent control tracts, the control pool is contaminated and the estimate is biased *up*. The firm-side null addresses where a student *goes*, not where they would have *offended*. Maybe donut design or “share of nearby tracts treated” column closes the threat.
- 2 Rambachan–Roth (2023) honest bounds on the pre-trend.** Connects to Comment 2: how much trend-slope can the post-period absorb before significance breaks? HonestDiD delivers the breakdown $\bar{M}$ in one run if $\bar{M}$ exceeds the slope implied by Table 2’s -0.711 property-crime cell, the pre-trend concern is laid to rest.
- 3 Deliver the public-safety externality in euros.** The abstract promises “sizable local public-safety externalities,” but never translates 20% into a per-trainee social-return ratio. Lochner–Moretti (2004), Bell–Costa–Machin (2022), Heller (2014) all close this loop. Offense reduction $\times$ Catalan cost-of-crime $\div$ marginal per-student DVT cost gives the externality *in euros*